

# Rafael Figueroa Gomez

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## PROFESSIONAL SUMMARY

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Electrical and Computer Engineering student at the University of Washington with hands-on experience designing and building embedded systems, analog circuits, and hardware prototypes for medical and energy applications. Proficient in Python, C++, Java, MATLAB/Simulink, KiCAD/Altium, and microcontroller-based development. Bilingual in English and Spanish with a strong track record of technical execution, cross-functional collaboration, and academic excellence.

## EDUCATION

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### University of Washington

Seattle, WA

*Bachelor of Science, Electrical and Computer Engineering — Expected June 2028*

### Whatcom Community College

Bellingham, WA

*Associate in Science Transfer Degree, Electrical and Computer Engineering. GPA: 3.95. June 2026*

Relevant Coursework: Advanced Multivariable Calculus, Electromagnetic Physics, Differential Equations, Linear Algebra, Java with Classes and Objects, Circuits, Engineering Numerical Methods/Computations

## EXPERIENCE

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### Coa Mexican Eatery

LaConner / Bellingham, WA 2022 – 2025

- Trained and cross-trained staff across 3 roles, reducing onboarding time by 2 weeks
- Led a team of 3 to deliver exceptional customer service, increasing sales by 5%
- Memorized full menu allergen profiles to ensure guest safety

### Peer Tutor

Whatcom Community College 2025 – 2026

- Guided 15+ peers each quarter through Calculus, Physics, Statics, and Differential Equations using clear, structured explanations
- Created a welcoming tutoring environment that reduced student anxiety around difficult coursework
- Applied subject expertise to break down unfamiliar problems for students in real time

### Peer Mentor

Whatcom Community College 2025 – 2026

- Organized and led engineering outreach events to encourage prospective students to pursue engineering
- Coordinated with team members to ensure events were inclusive and created a positive impression among new students

## PROJECTS

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### Negative Pressure Wound Therapy Device

- Developed an operational medical device to promote blood flow and accelerate healing, integrating Python, circuit analysis, and microcontroller-based software development
- Validated device through continuous multi-hour operation and presented findings to class; selected for encouragement to present at a state conference

### Heartbeat Monitor Circuit

- Built a prototype heartbeat monitor circuit incorporating analog filters, signal measurement, 3D modeling and printing, and IC design to produce an operational ASIC

### Solar Panel Energy Optimization (MATLAB/Simulink)

- Simulated multiple system parameters in MATLAB/Simulink to evaluate algorithmic approaches for solar panel maximum power point tracking, researching dynamic algorithm switching based on operating conditions
- Produced an IEEE-formatted research paper documenting methodology and findings

### Blackjack Game (Java)

- Implemented a fully functional Blackjack game in Java, applying object-oriented principles including classes, inheritance, and constructors

## SKILLS

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**Technical:** KiCAD/Altium, Soldering, 3D Modeling/Printing, Onshape, Raspberry Pi Pico (Microcontroller), Multimeter/Oscilloscope

**Programming:** Python, C++, MATLAB/Simulink, Java

**Languages:** English (Native), Spanish (Fluent)